



National Motor Freight Traffic Association, Inc.[®] Digital LTL Council

Product Requirements Document (PRD) for In-Transit Visibility

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Contents

Contents	2
1 Purpose	1-1
1.1 Document Conventions	1-1
2 Features and Deliverables	2-1
2.1 Requirements	2-1
2.2 Business Rules	2-1
2.3 Architecture & Implementation	2-2
2.4 Data Availability and Push Notifications	2-2
2.5 Associated Files	2-2
3 Use Cases, Definitions, & Workflows	3-1
3.1 Shipment Lookup Options	3-1
3.1.1 Verified Shipment Status Queries	3-1
3.1.2 Verified Shipment Data Response	3-2
3.1.3 Public Shipment Status Queries	3-5
3.1.4 Public Shipment Data Response	3-5
3.2 Standard Workflow Notification Definitions	3-12
3.2.1 Notification Types	3-12
3.2.2 Notification_DeliveryUpdate Workflow	3-12
3.2.3 Notification_DeliveryInformation	3-14
3.3 Standard Workflow Exceptions	3-15
3.3.1 Verified Endpoint & Non-Associated Shipments	3-16
3.3.2 Customer Pick-up / Drop off to Terminal Use Case	3-16
3.3.3 Interline Carrier Use Case	3-16
3.4 Reason Code Definitions	3-17
3.4.1 Delivery Exception Reason Code Definitions	3-17
3.4.2 Delay Reason Code Definitions	3-19
3.5 In Transit Field Standard Definitions	3-20
3.6 In-Transit Visibility Workflows	3-21
4 Release Criteria	Error! Bookmark not defined.
5 Proposed Timeline	3-1
5.1 Known Constraints	3-1
5.2 Assumptions and Dependencies	3-1
5.3 Out of Scope	3-1

1 Purpose

In-Transit Visibility will provide insight into a shipment's movement as it navigates through the pickup and delivery process. Once the carrier has picked up the shipment, shippers will be able to understand where the shipment is as it is in transit between pickup and delivery.

This In-Transit Visibility application programming interface (API) will include all status updates along the path of the shipment. It starts when the carrier acknowledges they have possession of a shipment and ends when the carrier delivers the shipment to the receiver and the receiver acknowledges receipt.

1.1 Document Conventions

The In-Transit Visibility process in this PRD is primarily outlined with a traditional happy-path, exception-free process whereas exceptions are outlined individually in their own workflow descriptions.

In addition, the following convention is used to specify type of event:

- **Verified In-Transit Visibility Events:** These events are the verified parties to the shipment through login or explicit addition by authorized account holder. They are highlighted yellow to indicate communication to tracking stakeholders and denotes.

Public Tracking Events: These events indicate what events and data are available to non-account owner parties and are highlighted in green.

Ideally, a Push notification option is the preferred method for status updates as well as a pull method available to obtain the most recent status update.

2 Features and Deliverables

2.1 Requirements

Project features and deliverables include:

Feature	Description
Provide insight into the pickup address/place of origin for the commodity.	Includes address, accessorial codes, limited access types, etc. affiliated with the pickup address.
Provide insight into the destination for the commodity.	Includes address, accessorial codes, limited access types, etc. affiliated with the destination address.
Provide insight into the terminal related to the shipment.	Includes address, contact information, and the status at the terminal.
Provide details on the commodity and any shipment specific attributes.	Includes handling unit specifics for the freight and shipment, such as count, type, weight, height, width, length, etc. See In Transit Field Standard Definitions for more information.
Provide insight into the statuses of the shipment through all phases.	Includes status and carrier acknowledgements as well as notifications. See Standard Workflow Notification Definitions and Standard Workflow Exceptions for more information.
Provide the ability to Search for In-Transit shipments.	Search criteria for In-Transit shipments should have the options available in Shipment Lookup Options to identify a shipment dependent on the role and availability of data points of the stakeholders.
Provide details for missed deliveries and provide reason codes.	Includes the responsible party for a missed delivery as well as the reason code that map to general descriptions. See Reason Code Definitions for more information.
Create transparency on Interline shipment status.	Adds Interline status, improving customer experience, and reducing confusion about interline shipments.

2.2 Business Rules

- In-Transit Visibility starts when the carrier acknowledges they have physical possession of a shipment.

- In-Transit Visibility ends when the carrier delivers the shipment to the receiver and the receiver acknowledges receipt.
- Notifications trigger a status update from the carrier at time of event and are sent to subscribed authorized parties affiliated with the shipment.
- Status updates are given throughout the path of the shipment. Status notifications grant visibility to stakeholders of the shipment to build trust between business partners through greater transparency, prevent phone calls into the carrier's call centers, and facilitate the stakeholder to self-serve support with better data available.
- Exception codes are initiated by the carrier when changes occur during the shipping that may affect delivery timeliness.
- Reason codes capture the responsible party for a missed delivery as well as allow a single, normalized reason code for shippers and 3PLs to map general descriptions while providing each LTL carrier the ability to expand specific definitions within a category to match their internal reason codes within a common structure.

2.3 Architecture & Implementation

In-Transit Visibility is the only LTL API with both Public and Verified options. This PRD and associated YAML are intended to provide guidance. Implementers are encouraged to use provided structure and data elements where possible, however, implementation decisions including endpoint structure and available data are at the implementer's discretion.

This PRD does not recommend unified or separate Public and Verified API endpoints for In-Transit Visibility.

2.4 Data Availability and Push Notifications

All data elements may not be available from all carriers. Data availability is dependent on carrier's existing systems. Carriers are strongly encouraged to supply as much data as possible, however, it is understood that this PRD is aspirational.

Users will note that some data elements are only available through push notifications. This is an intentional decision to limit the number of RESTful data elements and to encourage the use of push notifications.

2.5 Associated Files

This PRD includes an associated YAML file. The YAML contains detail including field types, required/optional status, nullable, and acceptable values.

3 Use Cases, Definitions, & Workflows

3.1 Shipment Lookup Options

The Search criteria for In-Transit shipments should have the following options available to identify a shipment dependent on the role and availability of data points of the stakeholders. Expected Payloads are different if the query is made from behind a login / verified account or a public inquiry. Payloads outlined are suggested best practices to simplify consumption by stakeholders.

3.1.1 Verified Shipment Status Queries

Query	Definition	Required Field(s)
Look up by Pro#	Returns results for a single shipment that is an exact match.	• Original carrier assigned Pro#
Look up by BOL#	Returns results for a single shipment that is an exact match.	• Shipper assigned Master BOL# • Origin Postal Code
Look up by PO#	Returns results for a single shipment that is an exact match.	• PO# on BOL • Destination Postal Code
Account # Query	Returns results for an array of shipments that match all criteria.	• Bill to Party Account number (Validated by login) • Starting date range of Pick-up • Ending date Range of Pick-up

3.1.2 Verified Shipment Data Response

Minimum required fields are listed for each notification; however, carriers are encouraged to offer as many of the optional fields as are available and appropriate to enhance the notifications. Alternatively, a separate API request can be made for the most current full shipment data available.

NOTES:

- See associated YAML for additional field details

Field Name	Description
ProNumber	The carrier assigned pro number associated with the shipment.
ReferenceNumbers	ReferenceNumbers schema
UniqueID	Pro + SCAC + eBOL Create Date (YYYYMMDD)
UpdatedPro	For interline or reconsignment.
PayType	Identifies terms of shipment (e.g., Prepaid or Collect).
ShipType	Identifies the payor of the shipment (Shipper, Consignee, or Third Party).
PickupAddress	Physical address the shipment was picked up from, using ShipmentStatusAddress schema
BillToAddress	Bill-to address, using ShipmentStatusAddress schema
BTaccountNumber	Account number for Bill To.
ConsigneeAddress	Physical address the shipment is consigned to, using ShipmentStatusAddress schema

Field Name	Description
DESaccessorial	An array to hold the list of services requested for the shipment. Valid values are listed in YAML
DESLimitedAccess	Limited access type from List tab [EBOL v2.1 Limited Access Types].
TimeCriticalType	Select Value from EBOL Standard Values on List tab.
TimeCriticalInfo	TimeCriticalInfo schema
GrossWeight	Weight in pounds of total shipment.
Commodities	ShipmentStatusVerifiedCommodity Schema
HandlingUnits	ShipmentStatusVerifiedHandlingUnit schema
ShipmentLocation	Shipment's current location using ShipmentStatusAddress schema
ShipmentMilestoneStatus	Indicates whether the status is Pick-up, In Transit, Interline, Out for Delivery, or Delivered.
ShipmentMilestoneStatusLong	Optional field for carriers to provide carrier-specific status information.
ShipmentTerminalInfo	Address and contact information for relevant terminal, ShipmentStatusVerifiedTerminalInfo schema
TimeZoneofShipment	The time in UTC (-X format) of current location.
UpdatedETA	The updated date, with or without time, that the shipment is expected to be delivered.
EstimatedStandardDay	The days in transit.

Field Name	Description
EstimatedDeliveryDate	The original date, with or without time, that the shipment was expected to be delivered.
OFDStopsAway	Indicates the number of stops away from the delivery destination, such as 3, 2, or 1 stop away. Only applicable when a shipment is Out for Delivery.
Reconsignment	Indicates if the shipment was reconsigned in transit.
ShipmentShortage	Indicates if product was lost in transit.
ShipmentDamage	Indicates if the product was damaged in transit.
AllShortorLostShipment	Indicates if this shipment is entirely lost in transit.
Late	Indicates if the shipment is going to be later than the original delivery date.
DelayReason	List of reason code on the List tab [Delay Reason Codes].
AppointmentRequired	Indicates if an appointment is required.
AppointmentStatus	Indicates an appointment status of Requested, Secured, or Missed.
AppointmentDate	The original date and time of the appointment.
PartnerCarrierName	The Interline or Agent Carrier's Name.
PartnerCarrierSCAC	The Interline or Agent Carrier's Standard Carrier Alpha Code® (SCAC)®
CrossBorder	Indicates if cross border services are required.
CrossBorderStatus	Indicates a status of In Bond, Hold, or Cleared Customs.

Field Name	Description
CrossBorderEntry#	The entry authorization number.
PODAvailable	Indicates if POD is available.

3.1.3 Public Shipment Status Queries

Query	Definition	Required Field(s)
Look up by Pro#	Returns results for a single shipment that is an exact match.	Original carrier assigned Pro#
Look up by BOL#	Returns results for a single shipment that is an exact match.	- Shipper assigned Master BOL# - Origin Postal Code
Look up by PO#	Returns results for a single shipment that is an exact match.	- PO# on BOL - Destination Postal Code

3.1.4 Public Shipment Data Response

NOTES:

- See associated YAML for field details

Field Name	Description
PRO	The carrier-assigned Pro number associated with the shipment.
Shipment Milestone Status	Picked up, In Transit, Interline, Out for Delivery, or Delivered.
Updated ETA	The updated date (with or without time) the shipment expected to be delivered.
Estimated Standard Day	Days in transit.

Schema Reference

ShipmentStatusReferenceNumbers

Field Name	Description
bolNumber	The eBOL or BOL number for the shipment
QuoteID	Quote ID provided by the carrier for the shipment
ShipmentID	Shipper-assigned reference number for the individual shipment (load #, shipment #, etc. This can also be the PRO.)

Field Name	Description
MasterBOL	Shipper reference number for pooled services.
poNumber	Purchase Order Number provided by Consignee. There can be multiple Purchase Order Numbers associated with a single shipment.
AdditionalReferences	Cross reference PRO provided by eBOL API
trailerNumber	Trailer number currently transporting the shipment.

ShipmentStatusAddress

Field Name	Description
addressName	The business or personal name for the address.
addressLine1	The primary address line for the address.
addressLine2	The secondary address line for the address.
City	The city for the address.

Field Name	Description
State	The two-letter state/province code for the address.
postalCode	The 5-digit + 4 or 5-digit for the United States, 5-digit for Mexico, or 6-character for Canada, postal code for the location.
Country	The 3-character country code for the given address.

ShipmentStatusVerifiedCommodity

Field Name	Description
Description	The description of the commodity.
Weight	The weight of the commodity.
weightUnit	The unit of weight of the commodity in kg or lbs.
Pieces	The number of pieces of the commodity.
PackagingType	PackagingTypes Schema

Field Name	Description
Classification	The freight class of the commodity.
nmfc	The National Motor Freight Classification® (NMFC)® code of the commodity.
nmfcSub	The NMFC sub code of the commodity.
Hazardous	Indicates whether the commodity is a hazardous material.
hazardousDescription	The description of the hazardous material commodity.
hazmatContactName	The name of who there should be contacted if there is a hazardous material-related issue with this commodity.
hazmatcontactPhoneNumber	The phone number for the contact person for hazardous materials-related issues with this commodity.
nestedOrStacked	Indicates whether the commodity is nested or stacked.

ShipmentStatusVerifiedHandlingUnit

Field Name	Description
Index	The index of the handling unit.
Count	The number of handling units containing the described freight.
Type	Type of handling unit.
tareWeight	Weight of packaging materials and handling units separate from the commodity weight.
Weight	The gross weight.
weightUnit	The unit of weight in kg or lbs.
Length	The length of the handling units being described.
Width	The width of the handling units being described.
Height	The height of handling units being described.
dimensionsUnit	Indicates the length unit of measure.
Stackable	Indicates whether the freight is stackable.

ShipmentStatusVerifiedTerminalInfo

Field Name	Description
TerminalID	The carrier-specific terminal ID.
addressLine1	The primary address line for the address.
addressLine2	The secondary address line for the address.
City	The city for the address.
State	The two-letter state/province code for the address.
postalCode	The 5-digit + 4 or 5-digit for the United States, 5-digit for Mexico, or 6-character for Canada, postal code for the location.
Country	The 3-character country code for the given address.
ContactName	The contact name for the terminal.
PhoneNumber	The phone number for the terminal's contact.
EmailAddress	The email address for the terminal's contact.

Field Name	Description
TerminalStatus	Indicates the status (arrived, unloaded, loaded, departed) at the terminal.

3.2 Standard Workflow Notification Definitions

3.2.1 Notification Types

There are two types of proactive shipment notifications: physical and informational.

Notification_DeliveryUpdate provides proactive alerts on a shipment's physical movements. It is triggered by:

- Changes in ShipmentMilestoneStatus
- Changes in terminalStatus
- Changes in countsToDelivery
- Other triggers at carrier's discretion

Notification_DeliveryInformation provides informational alerts not related to a shipment's shipmentMilestoneStatus. It is triggered by:

- Changes in appointments or accessorials
- Changes in customs information (including customs agent assigned or entry number provided)
- Reason codes assigned
- Other triggers at carrier's discretion

3.2.2 Notification_DeliveryUpdate Workflow

The In-Transit visibility begins with change of custody from the shipper to the carrier with necessary documents executed (BOL signed, E-BOL, or trailer swap) and the physical custody of the freight in the carrier's possession. The scenario used is a US domestic origin and destination without accessorial services required and serviced directly by a single carrier.

Notification	Definition
Departing Origin Notification	This message alerts stakeholders that the shipment has left the shipper's dock. Included in the message are dependent on the carrier's capabilities. Carriers that can send more information are always encouraged to do so.
Shipment Milestone Status changed to: Picked-up	Status changes to Picked-up when the shipment has been loaded on trailer.
Arrived at Terminal	Shipment has arrived at a terminal. Note: there will be multiple terminal status updates over the course of a shipment's lifecycle. Every LTL shipment passes through at least two terminals.
Unloaded at Terminal	Freight is stripped from the trailer and is on the dock of the terminal.
Loaded at Terminal	Shipment loaded on departing trailer to next terminal or out for delivery.
Departed Terminal	Trailer with shipment has left the terminal.
Shipment Milestone Status changed to: In-Transit	Status changes to In-Transit if it is en route to another terminal.
Shipment Milestone Status changed to: Out for Delivery	Status changes to Out for Delivery when shipment has departed final terminal en route to the customer. This status is reserved for final delivery to consignee not to an agent or interline partner.

Notification	Definition
Shipment Milestone Status changed to: Interline	Status changes to Interline when shipment is handed off to an Interline partner if the Interline partner is unable to provide status data. Status is to remain Interline until the shipment is delivered to the final consignee.
Stop Counts to Delivery:	Starting from within 3 Stops from Shipment Delivery Indicate the number of stops until that order delivery is scheduled. No timing estimates nor mileage only the number of stops. Update the count down from 3 to 2 to 1.
Shipment Milestone Status changed to: Delivered	Only updated with change of custody to final consignee, not upon transfer to Interline partner carrier.
POD Available	Indication from the carrier that an image of the POD is available for retrieval.

3.2.3 Notification_DeliveryInformation

The exceptions below are to trigger a status update from the carrier on occurrence to validated parties of the shipment. This list is not comprehensive.

Notification	Definition
Noted Late	When a shipment is expected to be delivered late by a day or more.
Noted Damage	When a shipment is noted to have damaged caused within transit.

Notification	Definition
Noted Short or Lost	When a shipment is noted to as missing elements or lost completely.
Noted Reweigh	When a shipment is noted as reweighed.
Noted Accessorial Added	When a shipment is noted as requiring additional accessorial services.
Reconsignment Notification	When a shipment is requested to be re-consigned.
Appointment Scheduled	When an appointment is scheduled.
Appointment Missed	When a scheduled appointment is missed. Include the reason codes and details of the code breakdown.
Appointment Rescheduled	When an appointment is rescheduled.
Missed Delivery	When a shipment failed to deliver as expected. See Delivery Exception Reason Code Definitions for information on the reason codes.
Customs Information Updated	When customs information is updated.

3.3 Standard Workflow Exceptions

Below are status updates for exceptions to the standard workflow scenario. Some may deviate from the standard workflow, while others are simple notifications of change or provide detailed reason codes. All proposed exception codes are initiated by the carrier when changes occur during shipping that may affect delivery timeliness. These status notifications provide shipment stakeholders with greater visibility, build trust between business partners through transparency, reduce calls into the carrier's call centers, and enable stakeholders to self-serve with better available data.

3.3.1 Verified Endpoint & Non-Associated Shipments

It is likely customers will use the In-Transit Visibility API for shipments not associated with their accounts. This may be due to a misbilled or miskeyed shipment, collect/3rd party shipment, related parties not listed on the Bill of Lading, or other causes.

If a customer's credentials do not have access to full Verified data, message 201 should be returned containing publicly available data. This will both provide high-level updates and notify the customer that they are not party to the shipment.

3.3.2 Customer Pick-up / Drop off to Terminal Use Case

If a customer picks up freight from a terminal the status is immediately moved to "Delivered". If a customer drops off freight at the terminal the status is immediately set to "In Transit". All other protocols and workflows apply.

3.3.3 Interline Carrier Use Case

If an Interline carrier is used, then partner carrier documentation is required. Lack of transparency on Interline/partner carrier shipments is a frequent cause of confusion for end customers.

The appropriate messages vary depending on the Interline carrier's data capabilities.

If the Interline carrier is **NOT** able to provide detailed status updates:

- Upon handoff to Interline carrier, **Shipment Milestone Status changed to: Interline**
- Shipment Milestone Status is to remain Interline until final consignee has custody of the shipment

If the Interline carrier is able to provide detailed status updates:

- Follow standard shipment notification path
- Shipment Milestone Status remains In Transit
- Add PartnerCarrierName, PartnerCarrierSCAC, and UpdatedPRO to Notification_DeliveryUpdate
- Include terminalStatus and Interline terminal details in notifications

3.4 Reason Code Definitions

3.4.1 Delivery Exception Reason Code Definitions

The table structure below captures the responsible party for a missed delivery as well as allowing a single normalized reason code for shippers and 3PLs to map general descriptions while providing each LTL carrier the ability to expand specific definitions within a category to match their internal reason codes within a common structure.

EXAMPLE:

310-2 Carrier Error	Reschedule	Carrier not able to service	310	2	Equipment Breakdown
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Responsible Party			Major Status Category				Digital Council Standard Messaging	Digital Council defined codes	Carrier defined	Carrier Defined sub codes for internal processing (example for reference only)
			100	200	300	400		x00-x90	0-999	
Carrier Action	Shipper Action	No Fault	Delivery Complete	Delivery w/ Exception	Reschedule	Closed Shipment	General Description	Basic Messaging Code	Sub Code	Carrier Detailed Description
		x	x				Success	100	1	Delivered Full and Complete
		x	x				Success	100	2	Delivered Full and Complete within Appointment
x				x			Delivery made with exception	200	1	Delivered with superficial damage
x				x			Delivery made with exception	200	2	Delivered with noted Product Damage
x				x			Delivery made with exception	200	3	Delivered Short
	x			x			Delivery made with exception	200	4	Waiting on Inspection
x					x		Carrier not able to service	310	1	Area not serviced today
x					x		Carrier not able to service	310	2	Equipment Breakdown
	x				x		Carrier not able to service	310	3	Customer request to reschedule
x					x		Carrier not able to service	310	4	Dispatch error
x					x		Carrier not able to service	310	5	Driver cut off
x					x		Carrier not able to service	310	6	End of day
x					x		Carrier not able to service	310	7	Delivery terminal capacity constraint
x					x		Carrier not able to service	310	8	Accident
	x				x		Carrier not able to service	310	9	Bad Address Data
	x				X		Carrier not able to service	310	10	Hold For Payment
	x				x		Appointment Issue	320	1	Need Contact Information
	x				x		Appointment Issue	320	2	Need Appointment Number
	x				x		Appointment Issue	320	3	Pending consolidation to full trailer
	x				x		Appointment Issue	320	4	Left message requesting appointment
		x			x		Weather	390	1	Inclement weather
		x			x		Civil Event	390	2	Civil Event
	x				x		Consignee Closed	330	1	Consignee closed - reschedule
	x				x		Consignee Closed	330	2	Consignee closed due to holiday (dispatcher)
	x				x		Consignee Closed	330	3	Consignee closed due to holiday (driver)
	x					x	Duplicate	450	1	Duplicate delivery (dispatcher)
	x					x	Duplicate	450	2	Duplicate delivery (driver)
	x					x	Delivery request cancelled	400	1	Consignee closed - cancel
	x					x	Delivery request cancelled	400	2	Customer request to hold at terminal
	x					x	Delivery request cancelled	400	3	Customer request to cancel
x						x	Carrier not able to service	410	1	Embargo/capacity constraint
x						x	Carrier not able to service	410	2	Equipment not available
x						x	Prohibited items	440	1	Prohibited or restricted freight (dispatcher)
x						x	Prohibited items	440	2	Prohibited or restricted freight (driver)
	x					x	Consignee Closed	430	1	Consignee closed due to strike (dispatcher)
	x					x	Consignee Closed	430	2	Consignee closed due to strike (driver)
	x					x	Unable to Access Consignee	460	1	Unable to access delivery location
	x					x	Unable to Access Consignee	460	2	Unable to contact Consignee for appointment
	x					x	Unable to Access Consignee	460	3	Refused Delivery

3.4.2 Delay Reason Code Definitions

NOTE:

List values and reason codes not listed below follow the latest published eBOL Standard.

Reason Code	Meaning	Customer Action
Weather		
Accident		
Breakdown		
Terminal Delay		
Needs Appointment		
Linehaul Delay		
Damaged		
Missing		
Need Customer Documents		
Equipment Breakdown		
Delay at Pickup		
Delay at Border/Customers		

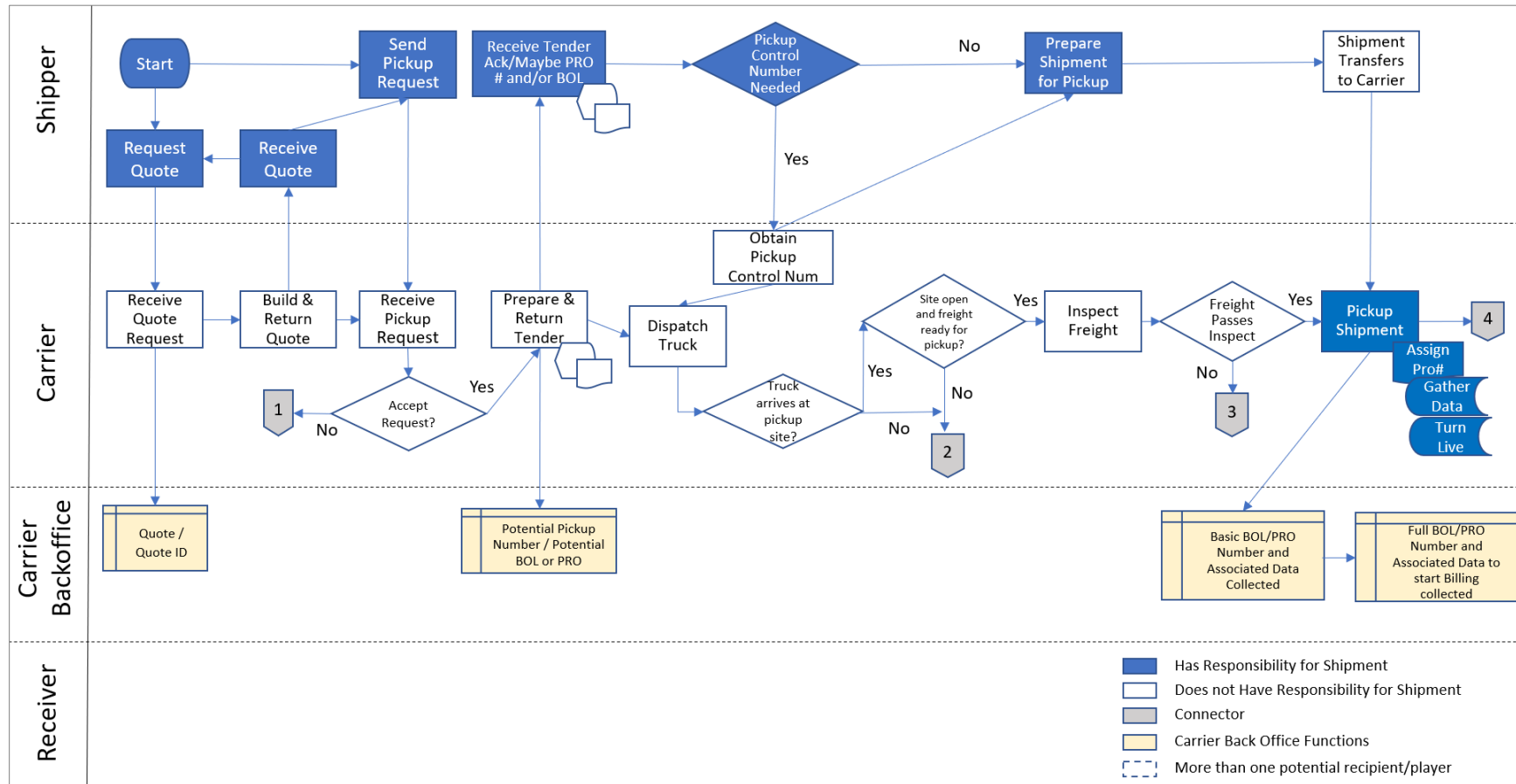
Reason Code	Meaning	Customer Action
Customer Requested Future Delivery		
Civil Event Related Delay		
Consignee Closed		

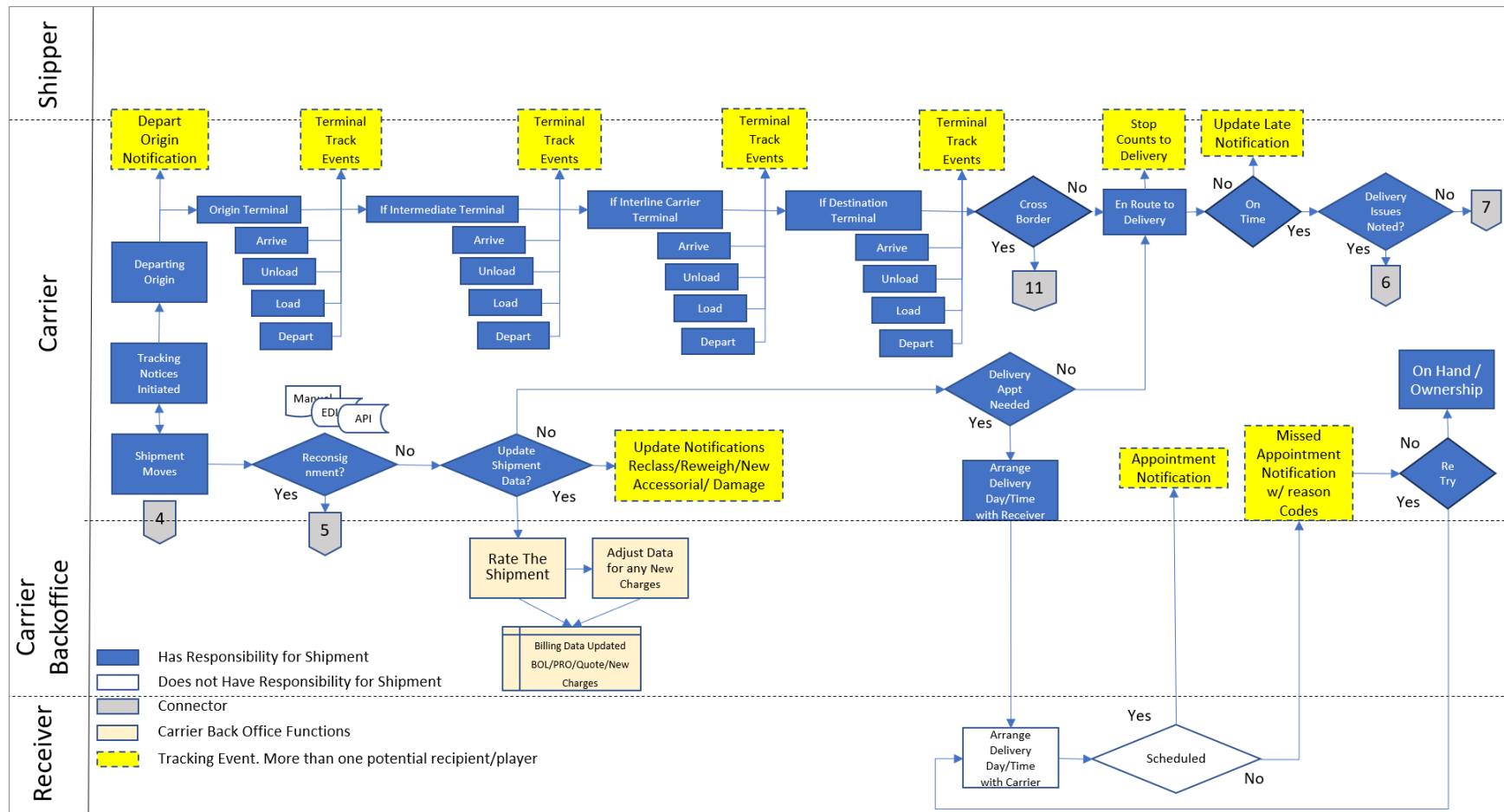
3.5 In Transit Field Standard Definitions

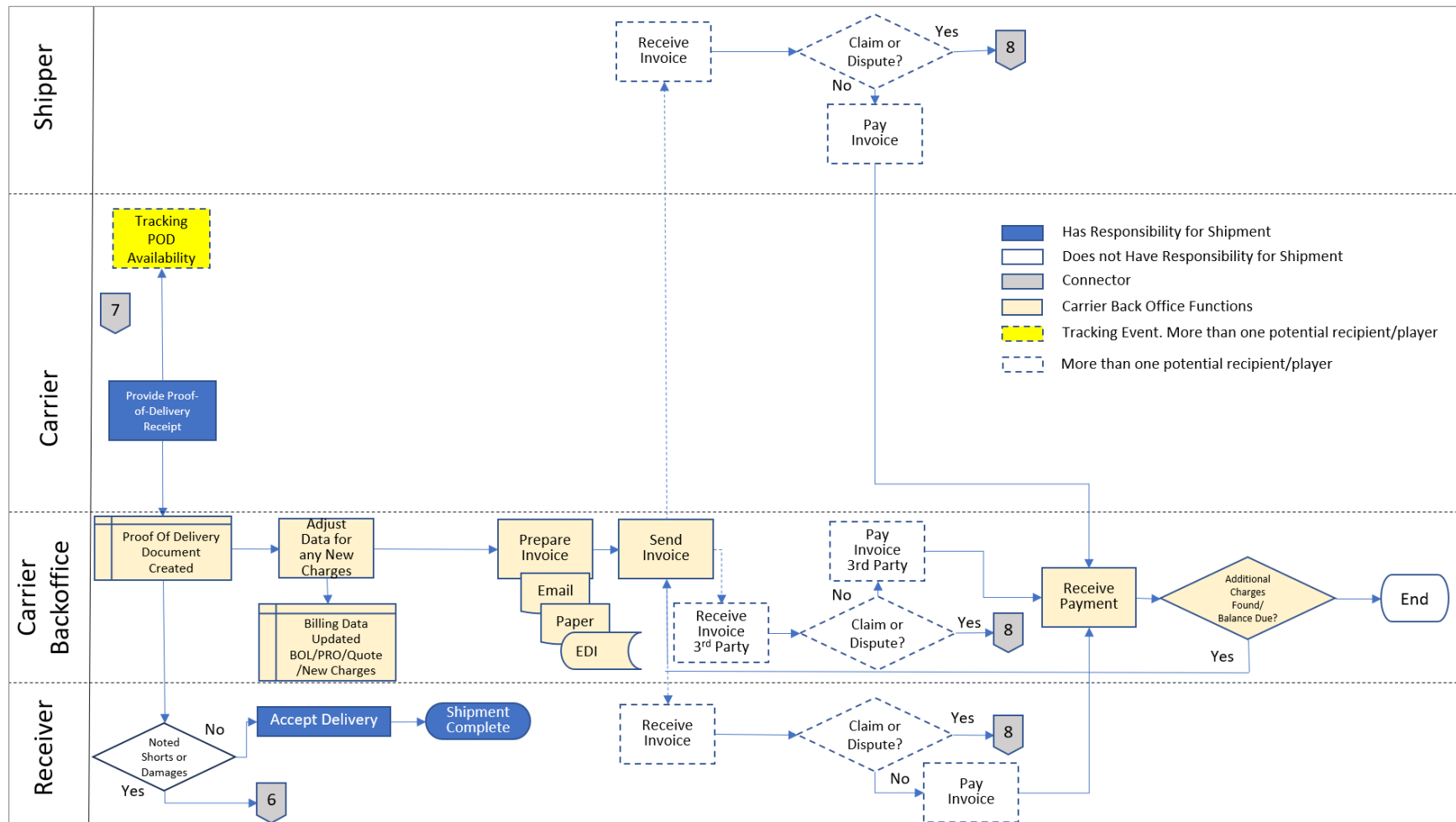
Refer to LTL-InTransitVisibilityAPI-v1.0.0.yaml for fields and definitions.

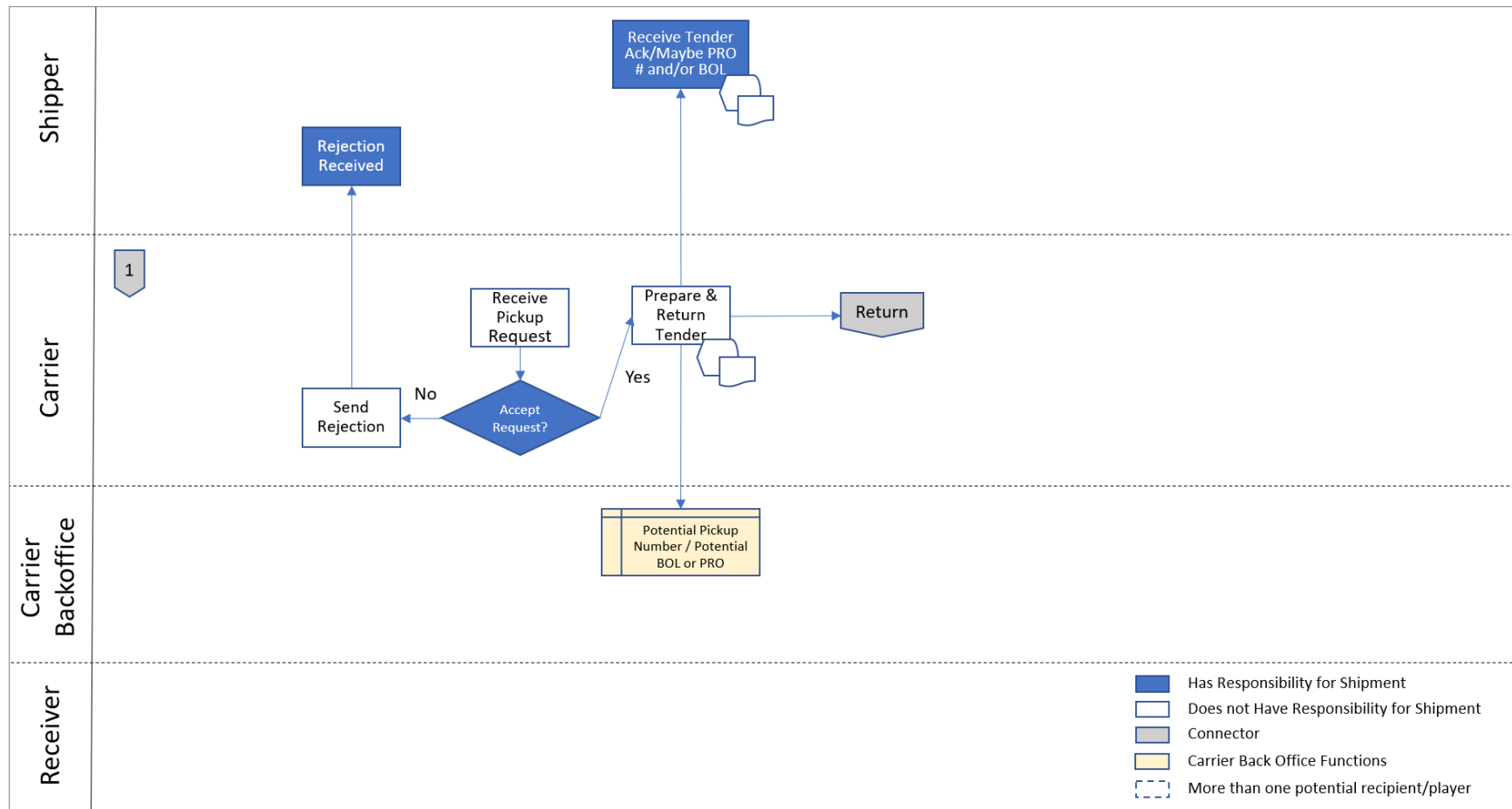
3.6 In-Transit Visibility Workflows

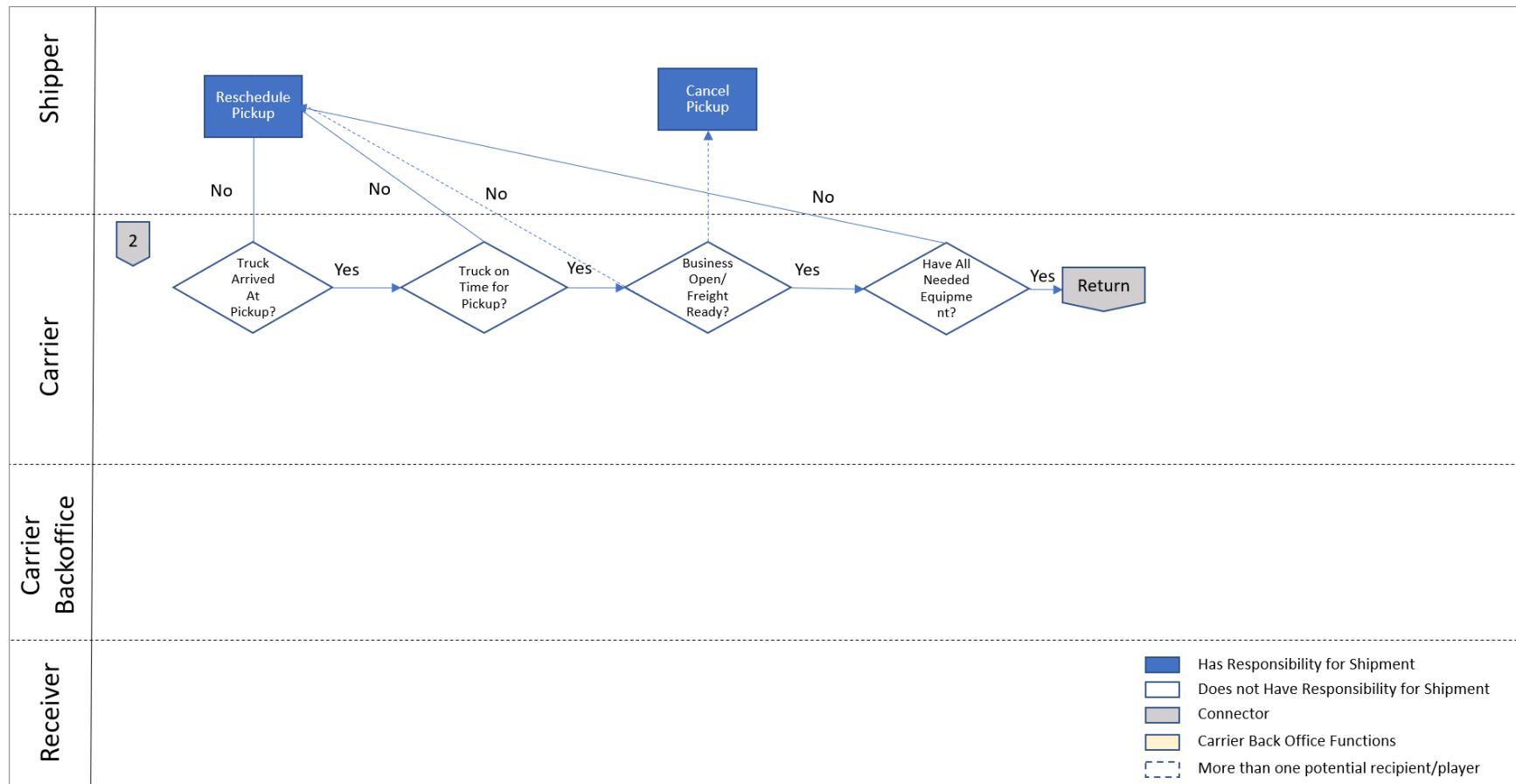
The following LTL workflows are generalized to represent the overall industry.

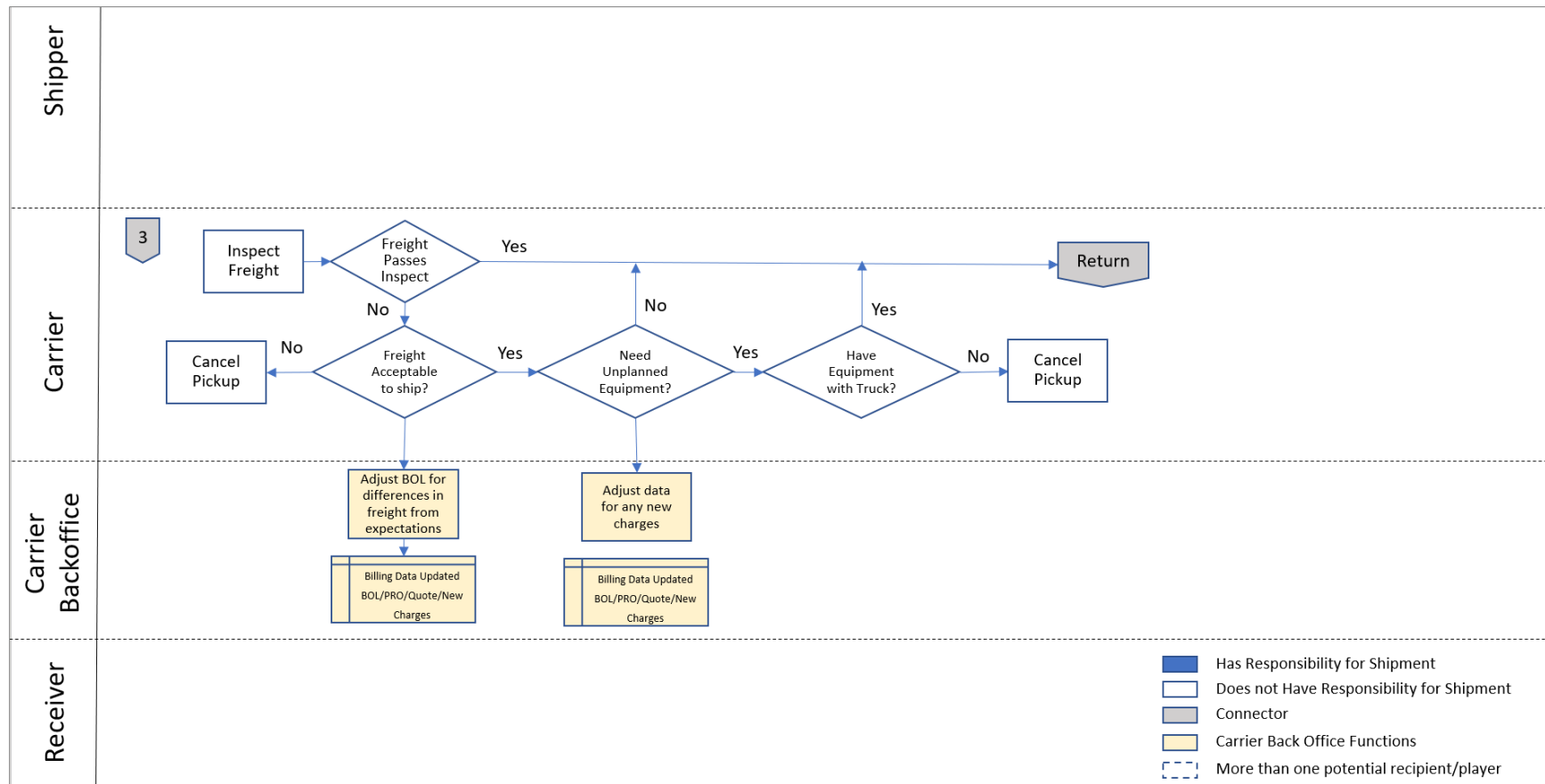


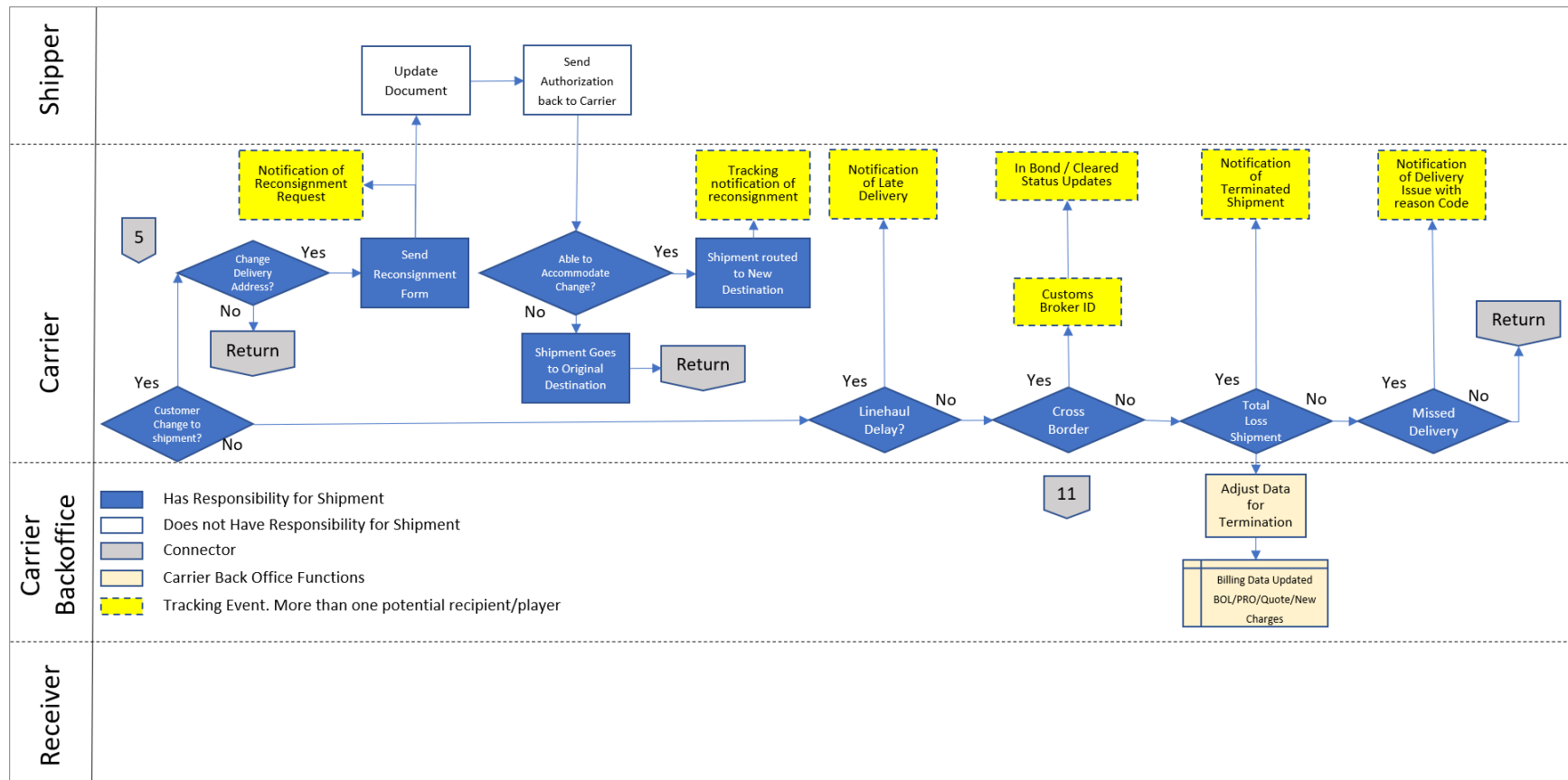


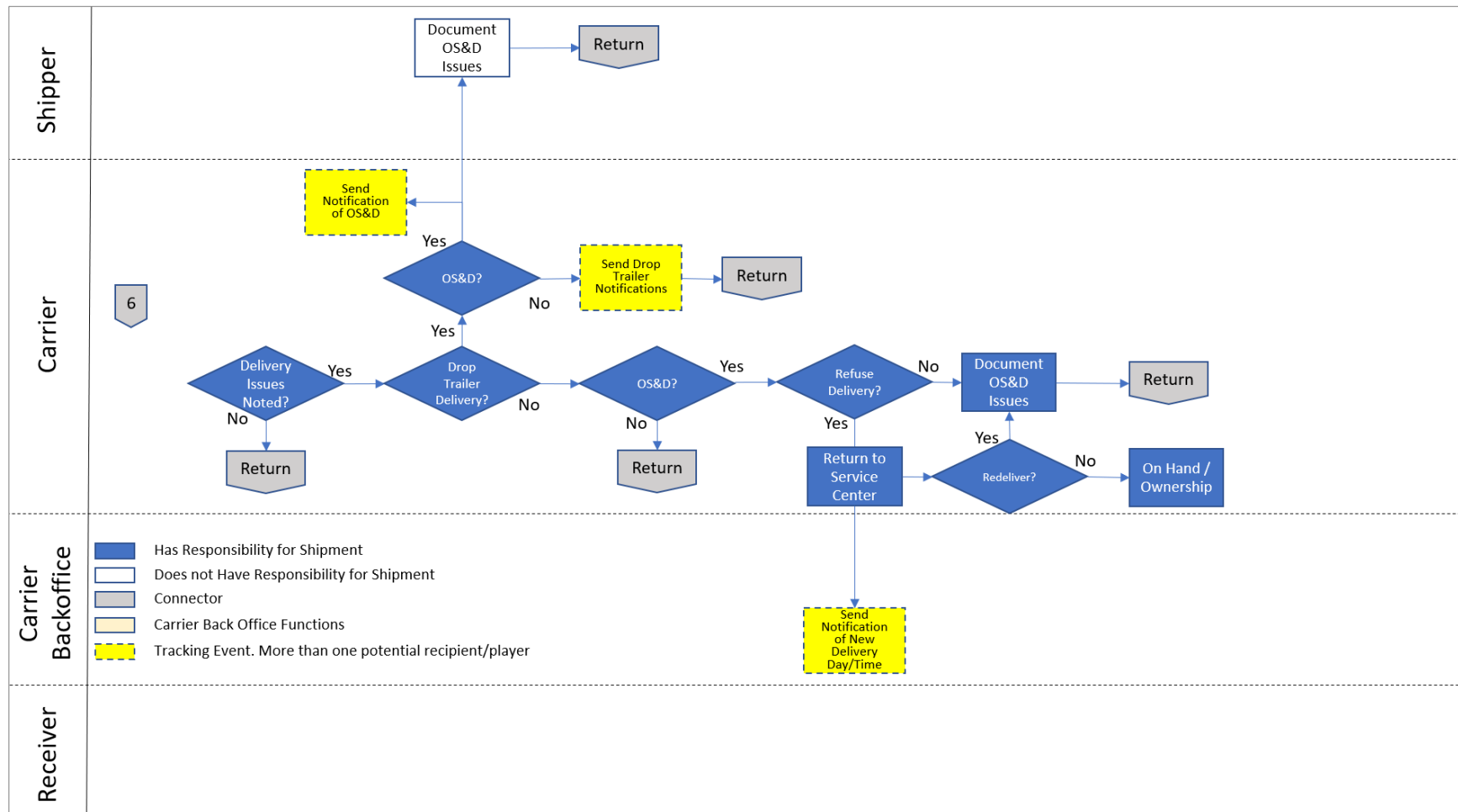


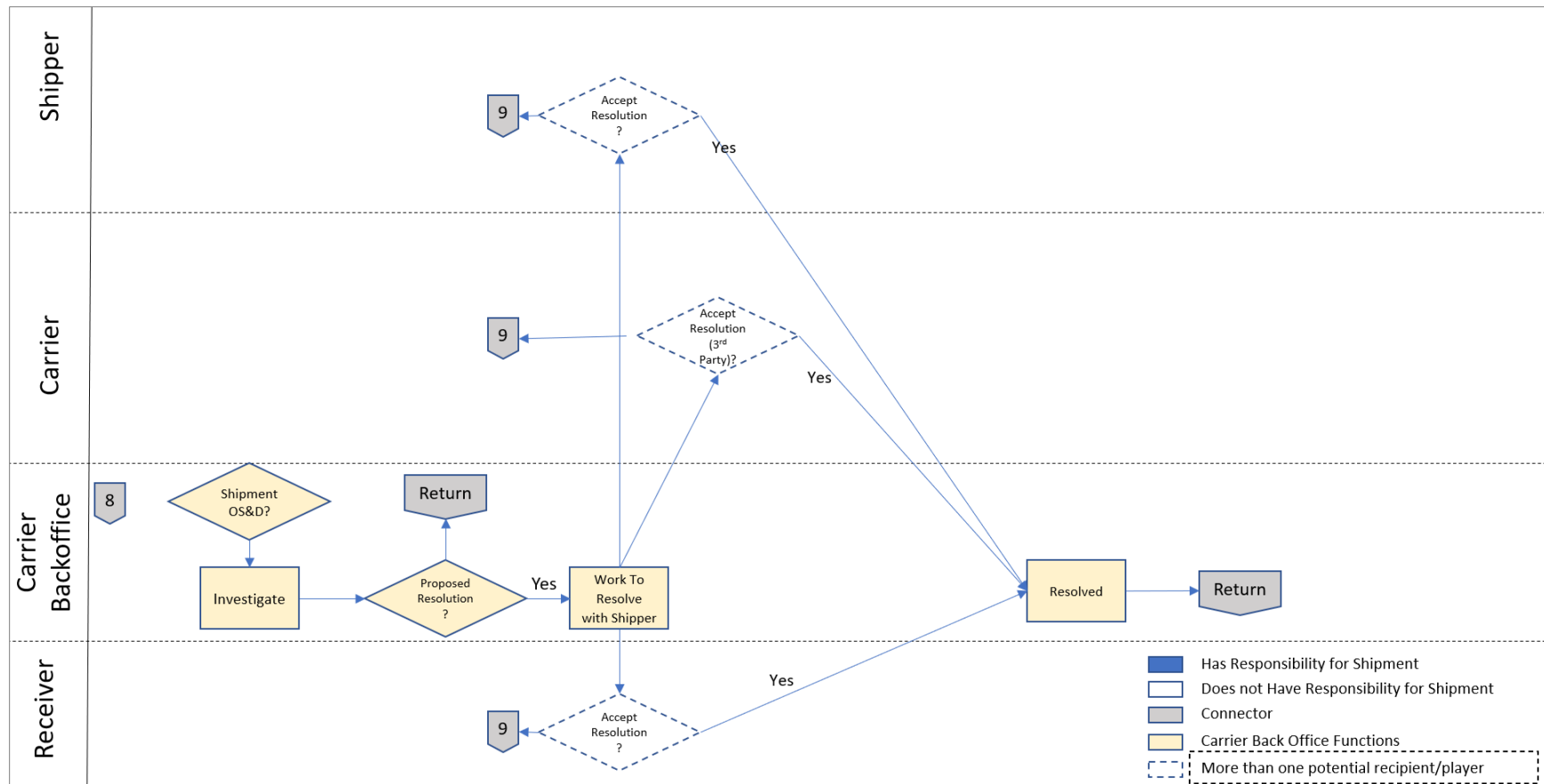


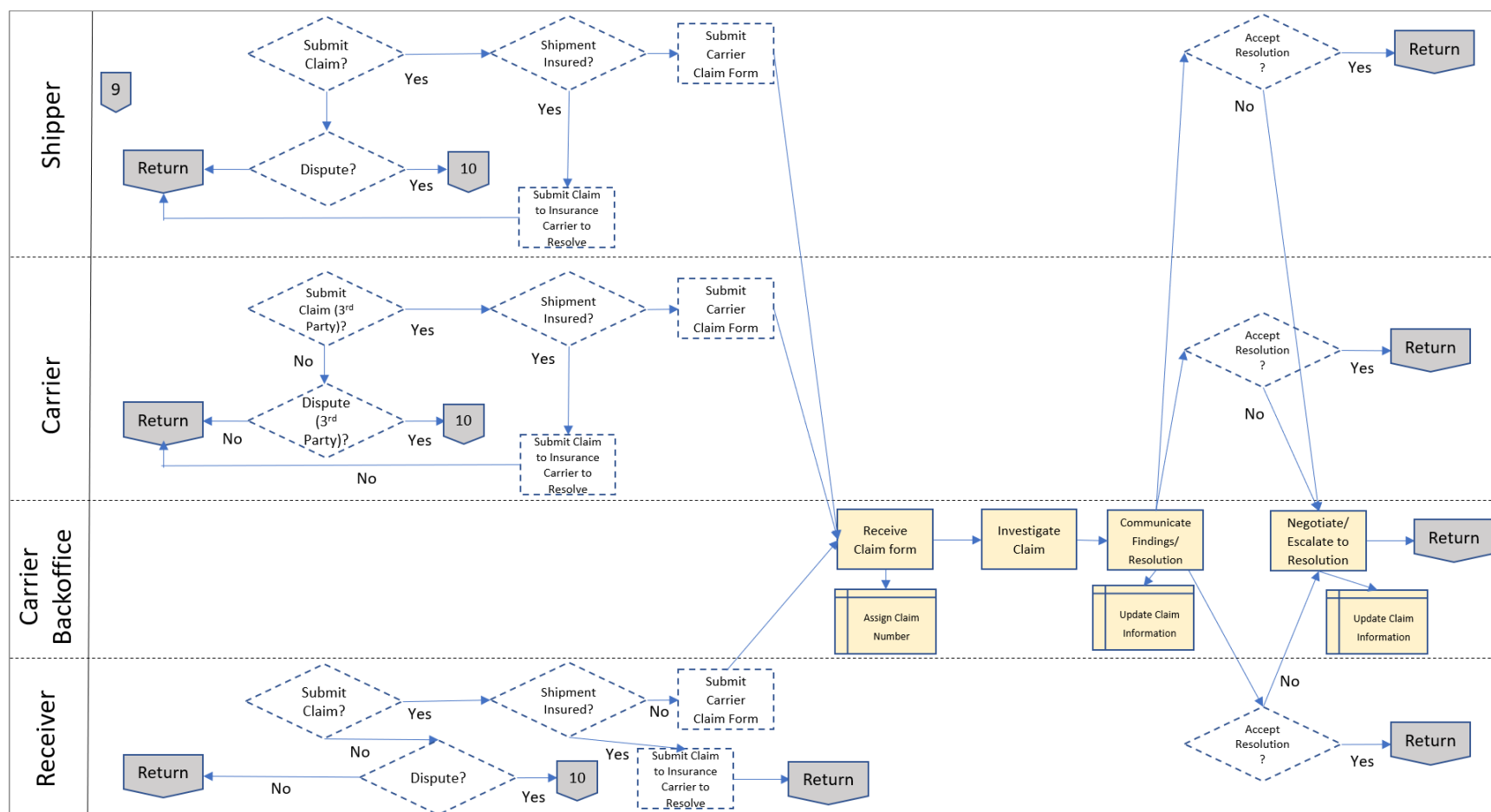


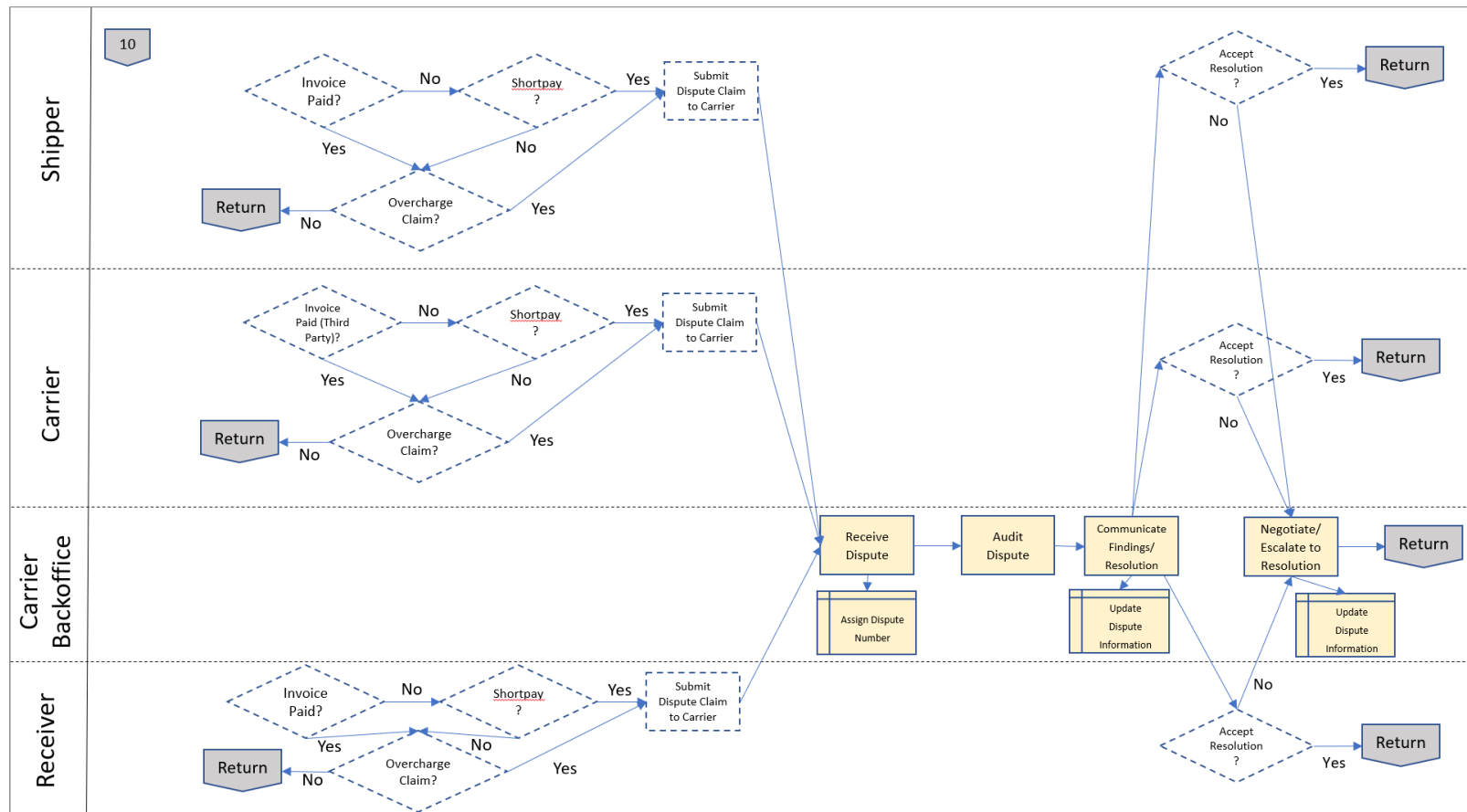












5 Proposed Timeline

Key milestones include:

Milestone/Deliverable	Timeline

5.1 Known Constraints

List any limiting factors that would pose a challenge to this project, such as staffing, cost, etc.

5.2 Assumptions and Dependencies

Provide a detailed list of assumptions (project factors believed to be true but that haven't been confirmed) to better determine the project's risk factors.

5.3 Out of Scope

- In-Transit Visibility begins when carrier takes possession of a shipment. As a result pickup appointment and other pickup accessorials are not in scope for this API.
- In-Transit Visibility notifications end when the Verified Party is notified that the Proof of Delivery signed by the Ultimate Consignee is available.
- The length of time for which Delivered status is available is at the carrier's discretion.